A380X standby instruments (ISIS)

The lower ISIS becomes the Standby Navigation Display (SND) with its power-up test, and the MODE pb swaps the SFD and the SND between the two ISIS, back to normal at each power-up

Aircraft	A380X
Where to find it	Centre instrument panel: the two ISIS (upper: SFD, lower: SND) and their MODE, MENU, LS/DIR TO, +/- and SET/SEL controls
Branch	`core/isis`, based on `master` (FlyByWire master 76db0aef1 merged in); independent.
Status	SND tested in the sim by the user: OK. SFD/SND reconfiguration tested by the user: OK. SND power-up test and the normal configuration at power-up: not sim-tested (unit tests).
Pictures	The SND instrument in the local browser harness, running the build of the branch, with ADIRU data typed in (the harness has no aircraft). No pictures for the power-up test and the reconfiguration.

1. Standby Navigation Display (SND)
2. SND power-up self-test
3. SFD/SND reconfiguration with the MODE pb
4. Normal ISIS configuration at each power-up

1. Standby Navigation Display (SND)

Aircraft: A380X | **Where:** Lower ISIS | **Status:** Tested in the sim by the user: OK. 10 unit tests.

What it is for

The A380 has two ISIS (integrated standby instrument systems): the upper one is the Standby Flight Display (SFD: attitude, speed, altitude), the lower one the **Standby Navigation Display** (SND): heading, position and a small navigation of its own, independent of the FMS (A380 FCOM DSC-34-10-20). On the FBW A380 the lower screen was black; it is now the SND.

The screen



Figure 1. The SND: ground speed, heading source IR3, heading rose with the yellow heading index and the green track diamond, aircraft symbol, position from ADIRU 3.

Item	Meaning
GS 370	ground speed, kt
IR3	the heading comes from ADIRU 3 (IR1 when ADIRU 3 has none); **TRU** in cyan in true reference
Heading rose	240° shown, a mark every 5°, numbers every 30°; white markers at 45° and 90°
Green diamond	the aircraft track
GPIR3 N 48 15.5 / E 003 32.3	the aircraft position (degrees and minutes) and its source
HDG, PPOS (amber)	heading or position not available

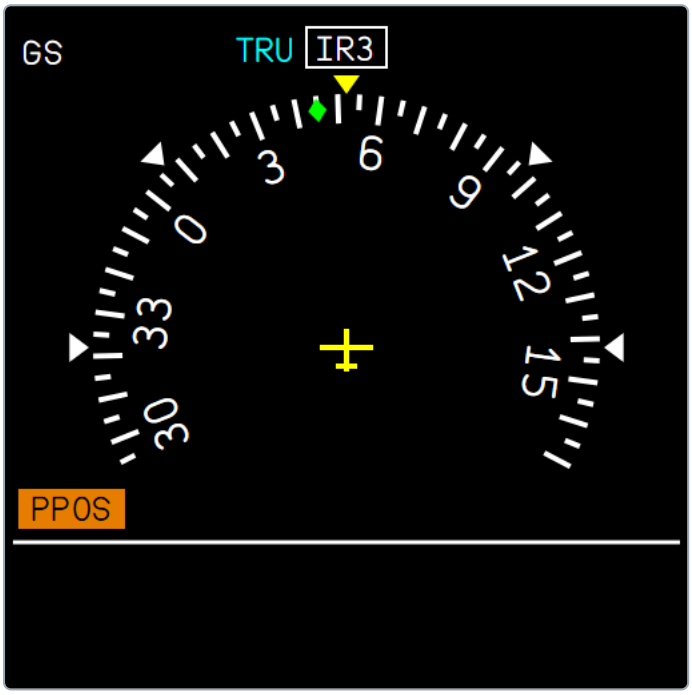


Figure 2. True reference (TRU), no position (PPOS), no ground speed.

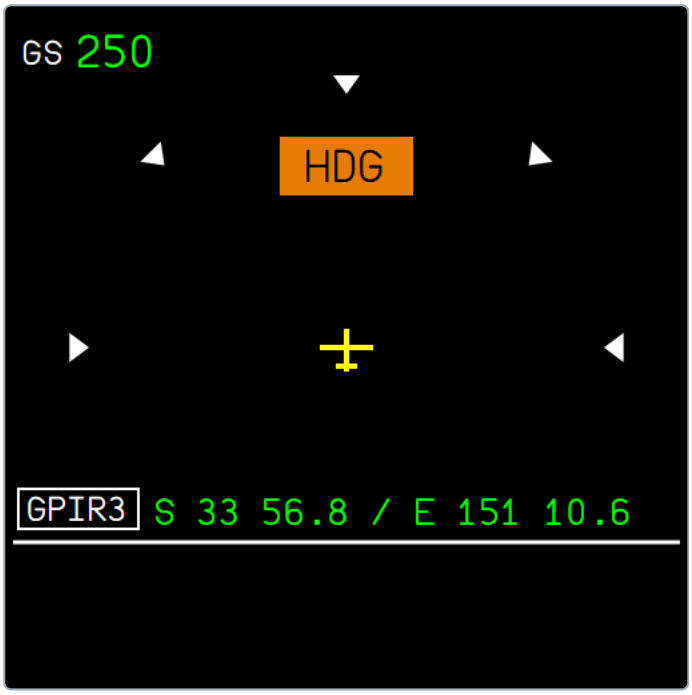


Figure 3. No heading from either ADIRU: HDG replaces the rose.

Buttons

Button	Action
MODE	SFD/SND reconfiguration: see the feature below
+ / -	brightness
MENU	opens the menu under the white line; press again to leave it
SET/SEL knob	turn: next item or value; push: select, or go to the next field
LS/DIR TO	DIR TO (see below); press again to leave it

The waypoint list

The SND has its own list of waypoints, entered by coordinates (it has no database). Press MENU: the menu shows INSERT WPT, INSERT FIX, EDIT WPT / FIX and CLEAR WPT / FIX in cyan; turn the knob to choose, push to select.

1. MENU, then push on **INSERT WPT**: the coordinates show, starting from the last waypoint (or your position).
2. Turn the knob to change the underlined field, push to go to the next one: N/S, degrees, minutes, tenths, then E/W, degrees, minutes, tenths.
3. Push on the last field: the waypoint is added at the end of the list.



Figure 4. The menu: INSERT WPT.



Figure 5. Entering the coordinates: the degrees of latitude being set.

EDIT and CLEAR WPT / FIX: turn to choose the waypoint ("2/3 N5015/E00232", then the FIX last), push to edit or to delete it.

DIR TO and the navigation

1. Press **LS/DIR TO**. With a list, turn to choose the waypoint and push. Without a list, enter the coordinates of the point as above.
2. The navigation starts from your position: FROM, TO (white) and NEXT under the line, the bearing and distance to the TO waypoint top right, and the course on the rose.
3. Passing abeam the TO waypoint, the SND goes on to the next one.



Figure 6. Two waypoints, navigation not active: the true bearing and distance between them.



Figure 7. After DIR TO: FROM / TO / NEXT, TO WPT 342° 126 NM, the course arrow.



Figure 8. About 5 NM right of the track: the deviation bar one large mark left of the aircraft (large marks every 5 NM, small every 2.5 NM).

The FIX

One point you want to watch (a beacon, an airport): MENU, **INSERT FIX**, enter its coordinates. The SND shows its distance top left, a magenta pointer towards it on the rose, and its coordinates in magenta at the bottom of the list. A new FIX replaces the old one; EDIT / CLEAR WPT / FIX offer it after the waypoints.



Figure 9. A FIX alone: 291 NM, bearing 060°.



Figure 10. The FIX together with a DIR TO: FROM / TO / NEXT / FIX.

Limitations and why

Limitation	Why
No automatic SFD/SND reconfiguration	FBW has no ISIS display failure; the MODE pb reconfigures by hand (see SFD/SND reconfiguration below)
The list is lost when the flight is reloaded	the ISIS keeps it while it runs

Position source always GPIR	the FBW ADIRS has no separate GPS / IRS positions yet
Coordinate entry, 10 waypoints at most, sequencing abeam	the FCOM figures of these procedures stop at the first frame: this is our reading

How to use

1. Read heading, track, ground speed and position on the lower ISIS.
2. Waypoints: **MENU**, **INSERT WPT**, set the coordinates with the SET/SEL knob (turn to change, push for the next field).
3. **LS/DIR TO**, choose a waypoint and push: the SND guides you there (TO WPT bearing and distance, course, deviation bar).
4. One point to watch: MENU, **INSERT FIX**. EDIT / CLEAR WPT / FIX to change the list.
5. **+ / -** for the brightness.

How to test

#	Do this	Expected result
1	A380X powered, ADIRS aligned (NAV). Look at the lower ISIS.	GS, IR3 (or IR1), the heading rose with the yellow heading index and the green track diamond, GPIR3 and the position at the bottom.
2	Press the **TRUE** north reference pushbutton (MAG / TRUE).	TRU in cyan; heading and track in true. Press again: back to magnetic.
3	Press **MENU**.	INSERT WPT, INSERT FIX, EDIT WPT / FIX, CLEAR WPT / FIX in cyan.
4	Push the SET/SEL knob on INSERT WPT; turn and push through the 8 fields.	The waypoint is added to the list under the white line.
5	Add a second waypoint the same way.	The true bearing and distance between the two show in the list.
6	Press **LS/DIR TO**, choose the first waypoint, push.	FROM / TO / NEXT, TO WPT bearing and distance top right, the course arrow on the rose.
7	Fly off the course by about 5 NM.	The deviation bar moves one large mark (5 NM) from the aircraft.
8	MENU, INSERT FIX, enter a point.	FIX distance top left, a magenta pointer towards it, its coordinates in magenta at the bottom.
9	Press **+** and **-**.	Brighter, dimmer.

2. SND power-up self-test

> **Aircraft:** A380X | **Where:** Lower ISIS | **Status:** Not sim-tested (2 unit tests). It was installed with the SFD/SND reconfiguration that the user tested, with no separate record.

What it is for

The two ISIS of the A380 are the same unit, SFD or SND (A380 FCOM DSC-34-10-20-10). The SFD (upper ISIS) already ran its power-up tests: a yellow INIT box and a countdown for 90 seconds. The SND came on at once. It now behaves as the SFD. The A380 FCOM gives no time: the 90 s and the 10 s below are the ones of the FBW SFD.

What you should see

Situation	What you should see
Cold and dark spawn, first power-up	INIT and the seconds left in a yellow box for 90 s; HDG and PPOS flags meanwhile; then the normal SND.
Spawn with the aircraft powered (runway, cruise...)	The SND is on at once, no test.
Power loss	The display stays on for 10 s; power back within 10 s: no new test. Longer: off, and the tests run again at the next power-up.
Power supply	As the SFD: DC ESS bus, or DC HOT 1 above 50 kt.

How to use

1. Nothing to do: after a cold and dark start, wait for the end of INIT (90 s) before using the SND.

How to test

#	Do this	Expected result
1	A380X cold and dark. Switch the batteries on (DC ESS powered).	The upper (SFD) and lower (SND) ISIS both show INIT with a countdown from 90.
2	Wait 90 seconds.	The SND shows its rose; HDG / PPOS flags until the ADIRS give heading and position.
3	Remove the power for less than 10 seconds, then restore it.	The SND stays on; no new INIT.
4	Remove the power for more than 10 seconds, then restore it.	The SND goes off, then runs INIT again for 90 s.
5	Spawn on a runway (powered).	The SND is on at once, without INIT.

3. SFD/SND reconfiguration with the MODE pb

> **Aircraft:** A380X | **Where:** MODE pb of either ISIS | **Status:** Tested in the sim by the user: OK. 2 unit tests.

What it is for

A380 FCOM DSC-34-10-20-20-10: the MODE pb "enables to switch from SFD to SND display or to switch off the corresponding standby display". Before, MODE only switched the SND off and on. Now each press goes to the next configuration, and one SFD is always displayed. Both ISIS screens carry both displays; the knob and the pushbuttons of an ISIS act on the display it shows.

What you see

MODE pressed	Upper ISIS (1)	Lower ISIS (2)
Normal	SFD	SND
1st press	SND	SFD
2nd press	SFD	off
3rd press	SFD	SND (back to normal)

- The SND waypoint list, DIR TO and FIX are the same whichever ISIS shows the SND.
- On the SFD, the **MENU** pb opens the BUGS page (the A380 FCOM puts the bugs in the SFD menu); the A320 ATT RST on that pb is gone (not in the A380 FCOM).

Limitations and why

Limitation	Why	What to do
No automatic reconfiguration.	The FCOM reconfigures automatically on an SFD display failure; FBW has no ISIS display failure.	Use the MODE pb.
A full MSFS restart is needed after an update.	The cockpit behaviour files and panel.cfg change.	Restart the sim after installing.

How to use

1. Press **MODE** on either ISIS to go to the next configuration (swapped, SND off, normal).
2. Use the knob and the pushbuttons of the ISIS that shows the display you want to work on.

How to test

#	Do this	Expected result
1	A380X powered, both ISIS after INIT. Upper ISIS: SFD, lower ISIS: SND.	Normal configuration.
2	Press **MODE** on the lower ISIS.	The upper ISIS shows the SND, the lower one the SFD.
3	Turn the knob of the upper ISIS (now SND), press its MENU.	The SND menu opens on the upper ISIS; the waypoint list is the one entered before.
4	Press MENU on the lower ISIS (now SFD).	The SFD BUGS page.
5	Press MODE again.	Upper ISIS SFD, lower ISIS off.
6	Press MODE again.	Back to normal: upper SFD, lower SND.

4. Normal ISIS configuration at each power-up

> **Aircraft:** A380X | **Where:** Both ISIS | **Status:** Not sim-tested (2 unit tests).

What it is for

The configuration chosen with the MODE pb survived a power cycle: an ISIS could come back swapped or off. The ISIS now start in the normal configuration (upper ISIS SFD, lower ISIS SND) at each power-up. The FCOM (DSC-34-10-20-20-10) says nothing about it: this is a design choice.

What you should see

Situation	What you should see
ISIS off, then powered (they run INIT)	Normal configuration, whatever MODE showed before.
Power loss shorter than the 10 s standby	The configuration is kept (not a power-up).
Spawn with the aircraft powered	The configuration is not changed by the spawn.

How to use

1. Nothing to do: after a power-up the ISIS are back to normal; use MODE again if you need another configuration.

How to test

#	Do this	Expected result

1	A380X powered, both ISIS on. Press MODE once.	Upper ISIS SND, lower ISIS SFD.
2	Remove the power for more than 10 seconds (batteries and external power off), then restore it.	Both ISIS run INIT; after it, upper SFD and lower SND (normal).
3	Press MODE once, then remove the power for less than 10 seconds and restore it.	The swapped configuration is kept.